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Hierarchical dual loop voltage and frequency control in stand alone microgrid with priority based intelligent load management

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ABSTRACT

The paper presents a hierarchical voltage and frequency control framework for peak and off-peak hours to enhance grid resiliency. The primary control is equipped with dual loop control mechanisms to maintain voltage and frequency during off-peak hours. Further, it provides additional inertia (virtual inertia) to reduce overshoot present in frequency transient (due to sudden switching of Loads). The secondary control strategy incorporates priority based intelligent management of Thermostatic Control Loads (TCLs) considering Customer Thermal Comfort (CTC). A Deep-learning based aggregation and control of TCLs (heating and cooling) is implemented to mitigate frequency deviation and moderates the overall energy usage during peak hours. Further, priority between the heating and cooling load (to be controlled) is decided with seasonal variation. Coordination between the primary and secondary control mechanism is established to enhance the dynamic response of the system during both peak and off-peak hours. Performance assessment of the proposed strategy is carried out in MATLAB/Simulink platform considering several dynamics. Further, the Hardware in Loop (HIL) is performed on OPAL-RT to showcase the efficacy of the proposed mechanism under realistic conditions.

1. Introduction

The increasing energy demand and advancement in converter technologies enable large integration of RES to the conventional grid. It provides several advantages such as the solution to the depletion of fossil fuels and carbon emission, however, its inconsistent nature and low inertia of stand-alone microgrid make it prone to several malicious events, that can render vulnerabilities in the system's performance [1]. Thus, controlling frequency and voltage in stand-alone microgrids is essential [2]. The PV inverter control strategy plays a major role in mitigating voltage and frequency deviation. The sequence-based voltage controller is presented in [3,4] where the authors have implemented an inverter controller to mitigate voltage unbalance. A distributed containment and consensus-based technique is discussed in [5] to restrict voltage levels (threshold) and regulate effective power-sharing (reactive) in an islanded AC microgrid. In [6] the distributed control for multi energy storage system for voltage regulation is incorporated. Here, the high bandwidth communication is required for coordination among DGs, it may result in delay, packet data loss, etc. However, above literatures [3–6] have not considered frequency control strategy which is essential during drastic load variations, and these papers also lacks peak hour management. In [7], model predictive control for frequency unbalance mitigation strategy is proposed considering battery SoC of electric vehicle. In [8] the load frequency control is presented to mitigate the frequency abnormalities. However, it lacks voltage and frequency control during peak hour or deficit generation condition. In [9] the frequency control scheme is implemented considering peak hour management, the paper did not address the voltage control scheme and lacks certain dynamics (frequency transient). Ref [10,11] proposed the droop-based frequency controller to enable power sharing and mitigate the frequency deviation. In [12] the novel frequency control is proposed, considering frequency transient. However, voltage control is not considered that may leads the system to brownout. The voltage and frequency control both is mandatory for stable operation of stand-alone microgrid. In [13], the authors have incorporated voltage and frequency control through battery unit in a stand-alone microgrid. Further, the battery SoC is maintained by the Diesel Generator (DZ) and that incurs fuel costs. In [14] a coordinated inverter fed V-f control strategy was introduced for stand-alone microgrid operation. Here, the PV control strategy coordinates with the battery control unit to enable power sharing among the micro-sources. However, they have not considered the issue of high overshoot (RoCoF) present in frequency transient

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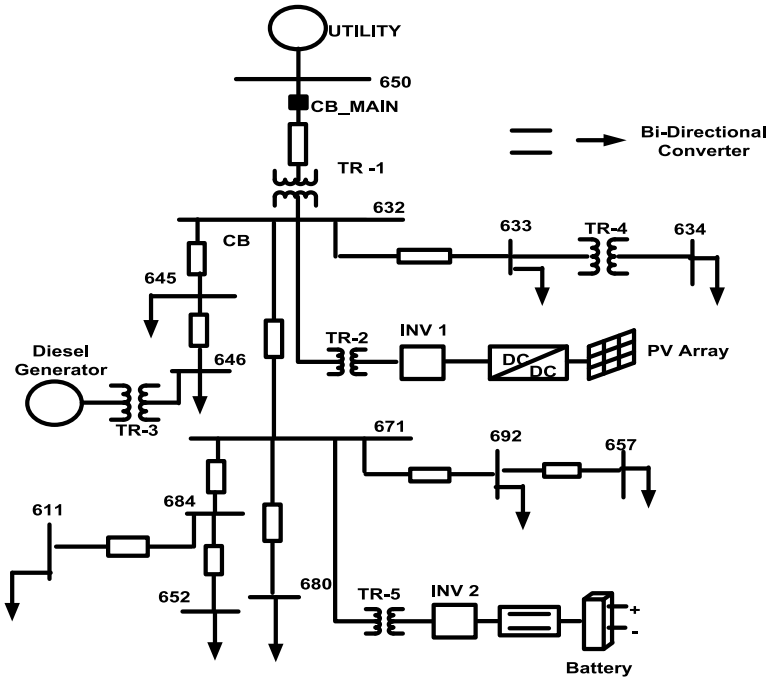


Fig 1. Modified IEEE 13 bus system under consideration.

Table 1
Microgrid specification.

PV details							
Unit	**Parallel Stacks**	**Series Strings**	**P_{max} (W)**	**V_{OC} (V)**	**V_{MPP} (V)**	**I_{MPP} (A)**	**I_{SC} (A)**
PV	105	56	175.11	29	23.6	7.42	8.1
Total Load							
Peak Load 1 MW							
Battery Details							
Nom. Voltage	**Series Strings**	**Parallel Stacks**			**Rated Cap per Cell**		
200 V	7	108			6.68 Ah		
Diesel Generator Specifications							
Base Power		**Peak Power**					
800 kVA		1 MVA					

occurs due to sudden load change. Thus, a complete control framework including voltage control and frequency control that able to minimize overshoot in frequency transient is required. Inverter controllers may fail to mitigate frequency mismatch during peak hours or during deficit generation and mandate load management. A strategic approach for managing TCLs' power consumption to monitor and minimize electricity demand at peak hours is presented in [15]. Ref [16] presented a voltage regulation strategy through scheduling the TCLs. However, fails to mitigate the frequency deviation. A Coordinated Frequency Control Strategy with TCLs management was proposed in [17], however, they have not discussed voltage control issue and high overshoot in frequency transient. Further, they have not mentioned clearly on CTC (Quality of comfort) while heat flow rate is high. The authors in [18] have framed a control method for voltage and frequency deviation with implementation of DSM. The inverter voltage control is done directly considering the three-phase parameters, thus that may inefficient during dynamic conditions and lead to brownout. Further, the frequency control strategy is not considered during high RoCoF (Load change is large).

The response of TCL control has an appreciable effect on frequency regulation as it consumes a significant amount of total load power. The above literature on load management had chosen either heating or cooling loads and this may not be effective in real scenario. Further, the manual aggregation of loads practically is difficult [18,19] and hence automation is required. The Artificial Neural Network-based TCLs aggregation is proposed in [20,21], however, they have not included inverter control mechanism for frequency regulation. In [22] the rule-based load management technique based on priority for residential loads is done. However, in the absence of an inverter controller, it is not economic to implement load management with small load switching that may lead to incur additional losses in terms of revenue. Implementing load management for small load change (or small frequency deviation) or during off-peak hours may put extra burden to the distribution company.

Thus, the paper incorporates the aforementioned gaps by implementing a hierarchical voltage and frequency controller:

1 Proposes a primary control technique to ascertain coordination among micro sources:
   (i) It implements a dual loop Inverter control strategy for standalone microgrid to compensate voltage and frequency deviation and provides virtual inertia to control the high overshot in frequency transient during sudden switching of loads.
   (ii) The pf and QV droop-based control strategy enable power sharing among the DGs and establish coordination between both the loops of primary control strategies.
2 Implements the secondary control strategy by intelligent load management technique:
   (i) It implements the Deep learning-based TCLs management considering priority between the TCLs (Heating or Cooling Loads first) during peak hours. The comparative performance analysis is also shown to showcase the superiority of the proposed method.
   (ii) It incorporates dynamic analysis of TCLs (cooling and heating loads) from a thermodynamics aspect considering the thermal model of a house with heat flow rate and structural parameters.

Further, the proposed controller is validated in real time environment with Hardware in Loop (HIL) to showcase its dynamic behavior.

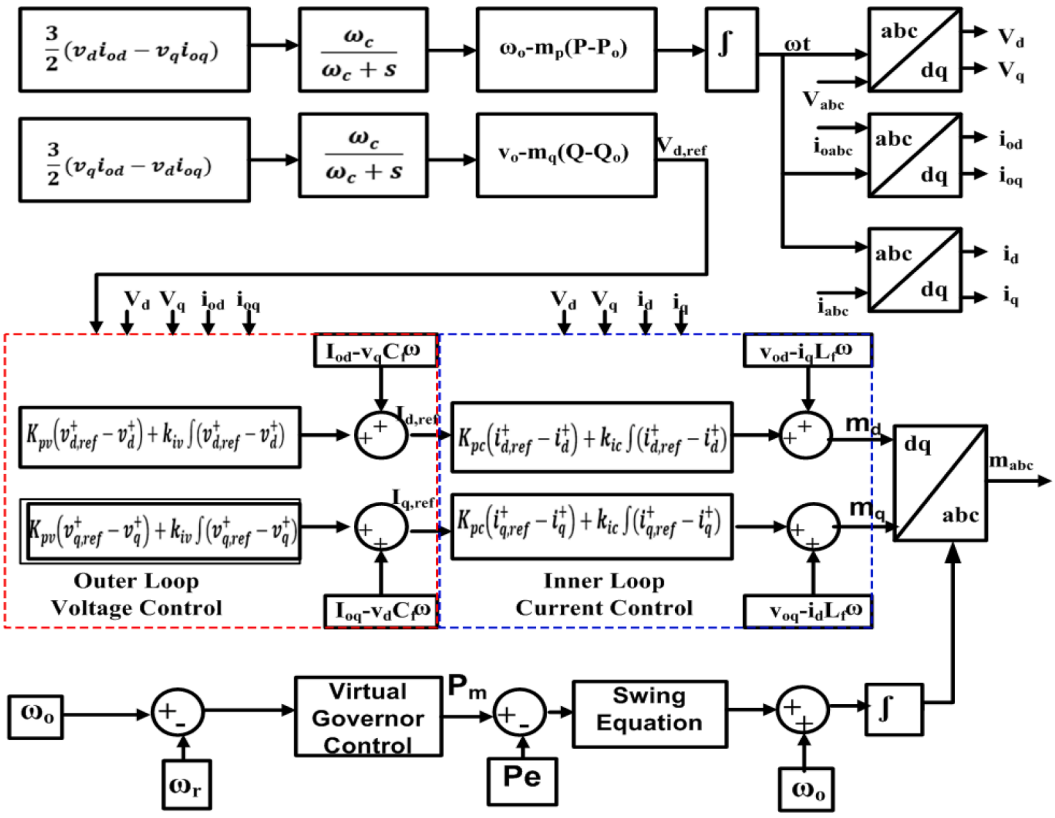


Fig. 2. Dual loop V-f control.

Table 2
Gain parameters of PI controllers and droop.

V controller(PI)				f controller(PI) Virtual governor		Droop gains			
K_{pv}	K_{iv}	K_{pc}	K_{ic}	K_p	K_i	$m_{p\ (PV)}$	$m_{q\ (PV)}$	$m_{p\ (Battery)}$	$m_{q\ (Battery)}$
0.72	217.5	34	1600	95×10^4	9,000,000	0.0000482	0.0003	0.0000464	0.00028

2. System under consideration

A modified IEEE 13 bus system has been considered for the study. It consists of micro-sources (PV, BESS, DZ) and TCL (25% heating and cooling loads) is considered to demonstrate the proposed concept shown in Fig. 1. It is assumed that the main grid is unavailable for the entire study. The details of the parameter of the studied system are shown in Table 1, where the line parameters are obtained in [23]. The base value of 1MVA taken into consideration with a rated voltage of 4.16 kV.

The generation transformers are of Yy0 type. In contrast, the distribution transformers are of Dyn11 type.

3. Hierarchical control strategy

The smooth operation of a stand-alone microgrid (low inertia) required an effective control strategy to maintain the voltage and frequency to its nominal value adhering to IEEE std 1547 [24]. Several voltage and frequency controller has been proposed [25], however, they have not been tested for the transient condition. Thus, the paper presented a hierarchical control strategy with a dual loop primary control strategy of pf and QV droop (shown in Fig. 2) based on cascaded voltage and current controller with a frequency control strategy to provide additional inertia. Further, an intelligent secondary control mechanism is incorporated by load management for peak hour operation.

3.1. Primary voltage control

In this technique, the voltage is controlled through cascaded voltage and current controller [26,27] where the voltage controller, controls the output voltage and provides a reference to the current controller that controls the current. A first-order low pass filter has been connected to measure the fundamental quantity. Further, the droop coefficient determines the variation of frequency and voltage for variation in power to be shared. The $V_{d,ref}$ (output voltage at particular loading) and m_q (Q-V droop coefficient) are related as given in Eq (1).

$$V_{d,ref} = V_n - m_q(Q - Q_n) \tag{1}$$

Where V_n is the nominal grid voltage and Q_n and Q are the rated capacity and instantaneous measured reactive power of DG.

The input of inner current controller is fed through outer voltage control loop as in Eqs. (2), (3).

$$i^+_{d,ref} = i^+_{od} - C_f\omega v^+_q + K_{pv}\left(v^+_{d,ref} - v^+_d\right) + k_{iv}\int\left(v^+_{d,ref} - v^+_d\right) \tag{2}$$

$$i^+_{q,ref} = i^+_{oq} + C_f\omega v^+_d + K_{pv}\left(v^+_{q,ref} - v^+_q\right) + k_{iv}\int\left(v^+_{q,ref} - v^+_q\right) \tag{3}$$

The final output is generated through inner current control unit as in Eqs. (4), (5)

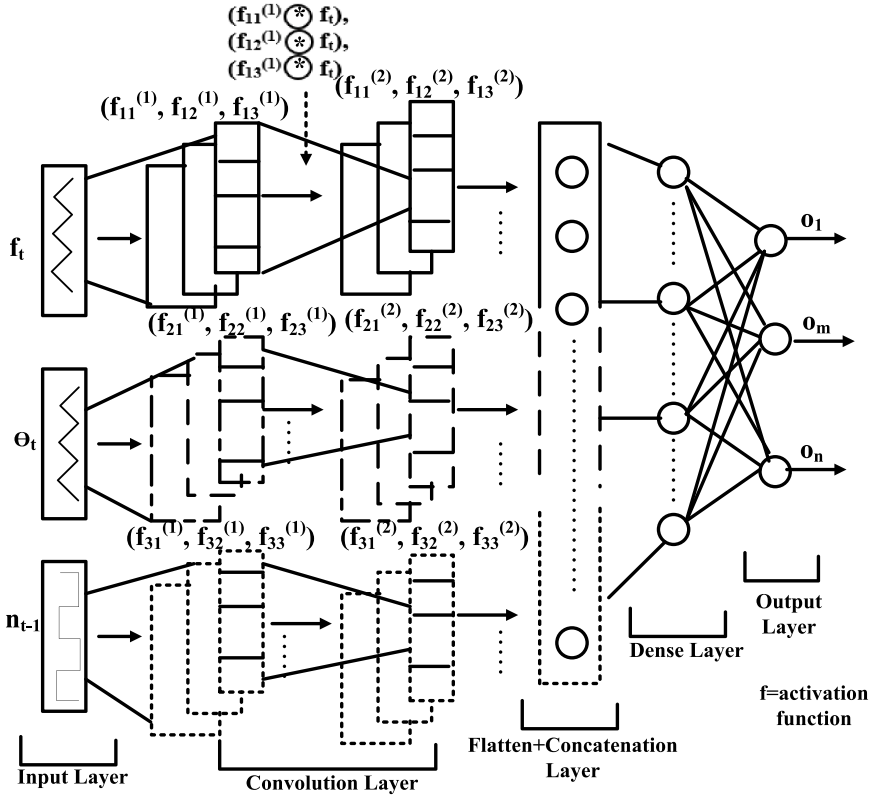


Fig. 3. CNN model for aggregation and control of TCLs.

Table 3
Architecture specification of TCLs.

Model	TCLs aggregation
No of Hidden layer	1 Dense, 2 CNN, 1(Concatenate+Flatten)
Neuron in CNN Layers	50, 25 Filter
Neuron in Dense Layer	2
Activation Function	Sigmoid
Percentage Dropout	20%
Pooling Type	Max Pooling
Loss Function	Mean Square Error
Optimization Function	Adam Optimizer with Back Propagation
No of Epochs	3500

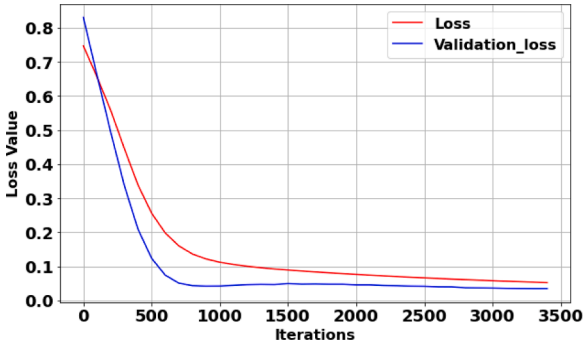


Fig. 4. Loss and validation loss of the model during training.

$$v_{ld,ref}^{+} = v_{od}^{+} - L_f \omega i_q^{+} + K_{pc}\left(i_{d,ref}^{+} - i_d^{+}\right) + k_{ic}\int\left(i_{d,ref}^{+} - i_d^{+}\right) \tag{4}$$

$$v_{lq,ref}^{+} = v_{oq}^{+} - L_f \omega i_d^{+} + K_{pc}\left(i_{q,ref}^{+} - i_q^{+}\right) + k_{ic}\int\left(i_{q,ref}^{+} - i_q^{+}\right) \tag{5}$$

The primary frequency control (virtual inertia based) is discussed in subsequent section.

3.2. Primary frequency control strategy

The sudden switching of load causes high change in RoCoF (introduces high overshot in frequency transient) in a low inertial microgrid, and it may lead to instability with absence of suitable control mechanism. Thus, frequency control strategy that able to provide additional virtual inertia support during sudden load switching is desirable and proposed in this paper. The active power control framework is proposed to mitigate the frequency disturbance [12]. The control mechanism injects active power that is proportional to the frequency unbalance. Further, the droop-based mechanism is incorporated for power sharing among the DGs. The relation between measured frequency (ω^*) and p-f droop coefficient (m_p) is as given in Eq. (6).

$$\omega^* = \omega_n - m_p(P - P_n) \tag{6}$$

Where ω_n is the nominal grid frequency, P_n is the rated capacity of DG and P is the measured active power. Further, the angle obtained from the droop control is given as:

$$\theta = \int \omega(t)dt = \int\left[\omega_o - m_p\left\{\mathscr{L}^{-1}(\omega_c P_{abc}(s)\,/\,(s+\omega_c))\right\}\right]dt \tag{7}$$

Where the ω_c is the cut-off frequency of the First-Order Low-Pass Filter and P_{abc} is the unfiltered three phase active power. The pf droop-based frequency controller is further equipped with swing equation to restrict the large deviation in RoCoF during abnormal load switching by

Table 4
Performance comparison of proposed CNN based classifier with kNN and SVM.

	Training Iteration	Accuracy (%)	Precision (%)	Testing accuracy Class 1 (%)	Class 2 (%)	Class 3 (%)	Overall (%)
CNN	3500	96.7	94.9	99.2	99.1	98.9	99.06
kNN	1600	89.7	86.52	92.4	91.2	89.8	91.13
SVM	1800	93.5	93.98	98.41	96.06	89.87	94.78

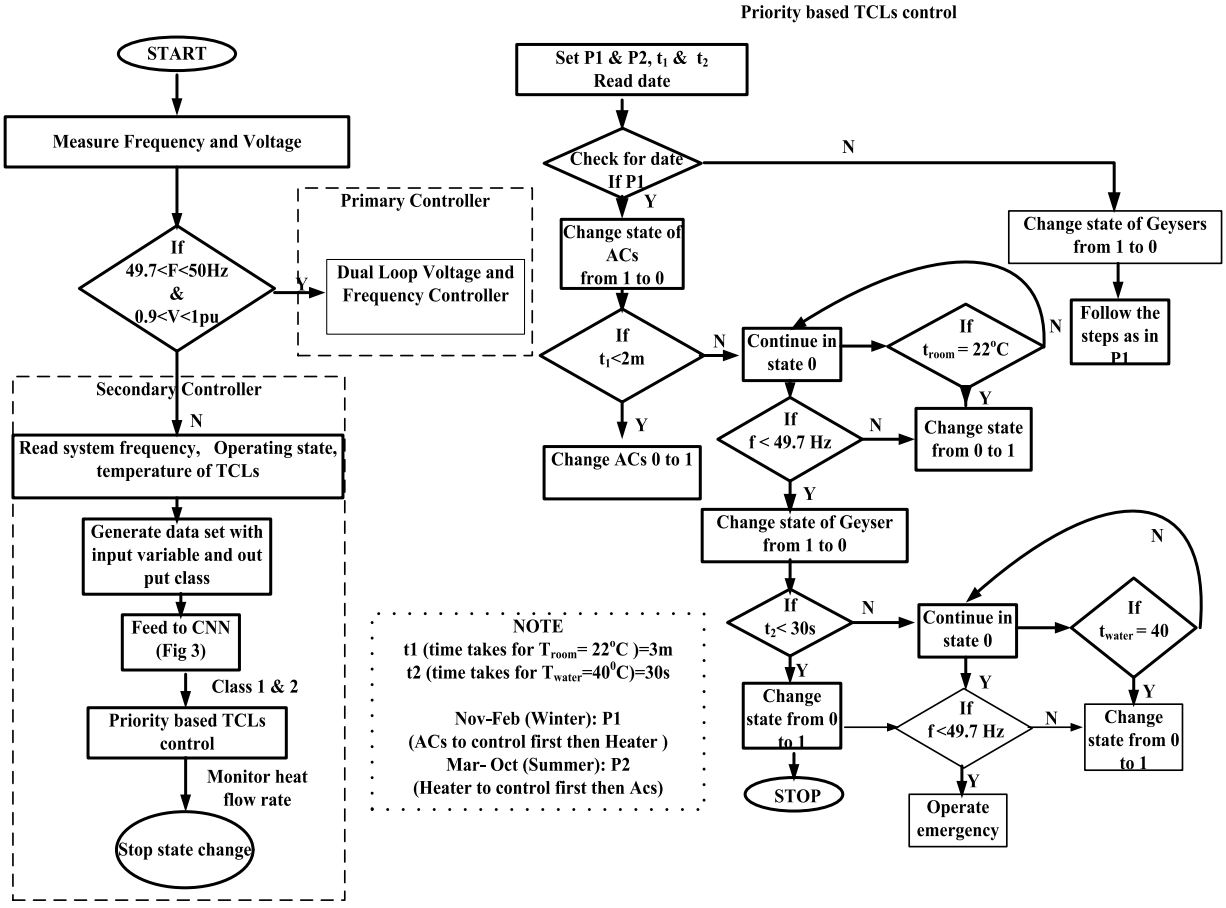


Fig. 5. Operational flow diagram of proposed hierarchical control strategy.

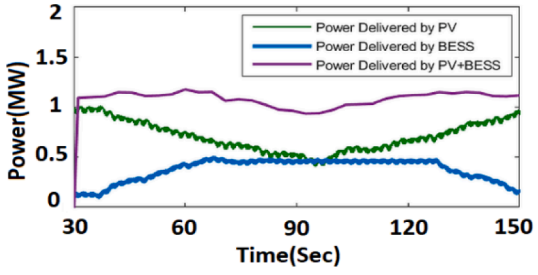


Fig. 6. Pf and QV droop-based power sharing among DGs.

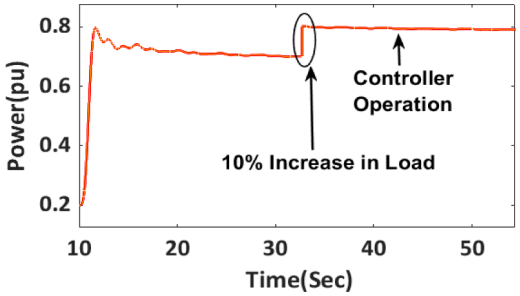


Fig. 7. Load profile at bus 645.

providing additional virtual inertia through governor action [10]. The integrator is used to make the response free from offset. The formulae of pf droop-based frequency controller equipped with swing equation is depicted in Eqs. (8)–(13). The basic equation of motion is given in Eq. (8).

$$J\frac{d\omega_m}{dt} = T_m - T_e = D\omega_m = T_a \tag{8}$$

The dynamics of frequency is represented in Eq. (9).

$$P_m - P_e = \mathrm{D}\Delta\omega + \mathrm{J}\omega_n\frac{d\Delta\omega}{dt} \tag{9}$$

Where J is the moment of inertia, ω_r is the electrical angular velocity, P_m and P_e are the mechanical and electrical power respectively.

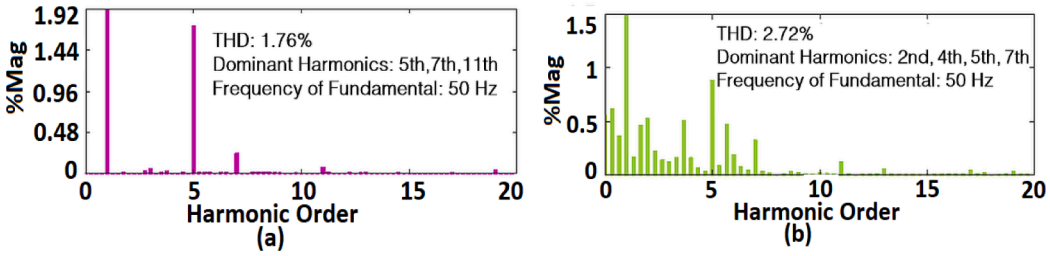


Fig. 8. (a) Voltage THD (b) Current THD.

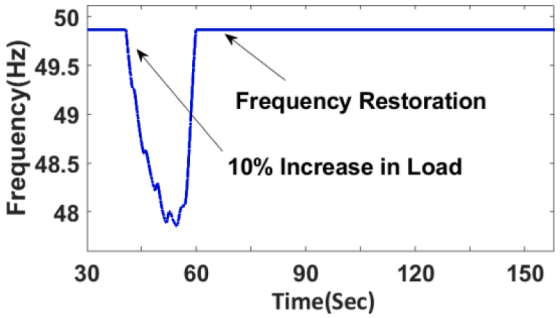


Fig. 9. Frequency response without inertia control.

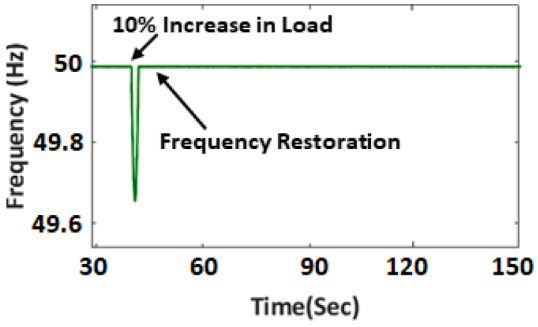


Fig. 10. Frequency response with inertia control.

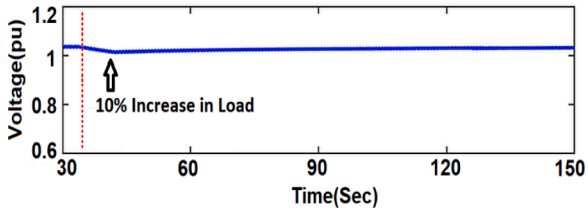


Fig. 11. Voltage waveform during load change at bus 632.

Eqs. (10) and (11) is obtained after combining the conventional droop and swing equation.

$$J\omega_r\frac{d\omega_r}{dt} = P_0 + m_p(\omega_0 - \omega_r) - P_e \tag{10}$$

$$\text{Where, } P_m = P_0 + m_p(\omega_0 - \omega_r) \tag{11}$$

$$\text{Swing equation is, } (J\omega_o + D)(\omega_o - \omega_r) \tag{12}$$

Therefore, the equations regarding the inertia-based frequency control is given below

$$\omega = \omega_o + \Delta\omega \tag{13}$$

Where, $\Delta\omega = \frac{m}{J\omega_o}$

$$m = -P_e - D\Delta\omega + K_p\Delta\omega + K_i\int \Delta\omega\, dt$$

Where P_0 is the nominal power, m_p is the droop constant, ω_0 and ω_r are the nominal and electrical angular frequency. The combined model of V-f control maintains the voltage and frequency adhere to IEEE 1547.1 at any change in load during off-peak hours (while generation and demand mismatch is low). The moment of inertia (J) is 174 kgm^2 and Damping constant (D) is 0.55 The LCL filter is used to mitigate the harmonic content from the output with $L_{f1} = 0.49$ mH, $L_{f2} = 0.32$ mH and $C_f = 91.5$ μF. The gain value of Droop and PI controller is depicted in Table 2. The peak hours issues can be alleviated through secondary control technique of load management.

3.3. Secondary control of load management

The primary control strategy may fail while generation demand mismatch is high (in peak hour, deficit generation condition etc.). These situations can be handled with load management. Several literatures are available on load management, however that are either adopted manual aggregation or not mentioned clearly the methods adopted for load management out of a large population of loads. Further, the manual aggregation of large number of loads (TCLs) is quite difficult and inefficient to meet the purpose. Thus, an automatic aggregation is mandatory. Further, registration of newly connected schedulable loads is essential. The work aims with developing the deep learning-based aggregation and selection of TCLs, and priority-based control of loads. Further its intelligence allows it for auto updating in regular interval. The management of TCLs (heating and cooling) while frequency reaches 49.7 Hz was carried out in four steps such as Aggregation of TCLs to participate in management, Verification of operating temperature of aggregated TCLs, Segregation of aggregated TCLs (heating and cooling) based on temperature band (16–22 °C for ACs and 40–50 °C for Geyser) issue of priority-based control signal to the segregated TCLs.

3.3.1. Intelligent load management

The large power consuming loads such as ACs and Geysers are considered for management to restore frequency. Further, these TCLs loads are basically employed to provides comfort to end users. Thus, load management is carries out considering CTC and weather condition. The ACs will manage first in winter and then geysers as per requirement and vice-versa for geysers in summer. The operation of TCLs from the thermodynamics point of view [18] is demonstrated through Eqs. (14)–(18). Assuming dx to be the thickness of an arbitrary room's wall which is exposed to the surrounding, of temperature T_{out}. The temperature of the inner part of the wall may be denoted by T_{in}. If the heat-flow through radiation is neglected, the energy balance on a room's wall can be expressed as.

$$q_i = \pm\frac{(T_{in} - T_{out})}{\Re_{wall}} \pm q_{source} \tag{14}$$

Where q_i is the heat inside the room, $\Re_{wall}$is the wall's thermal

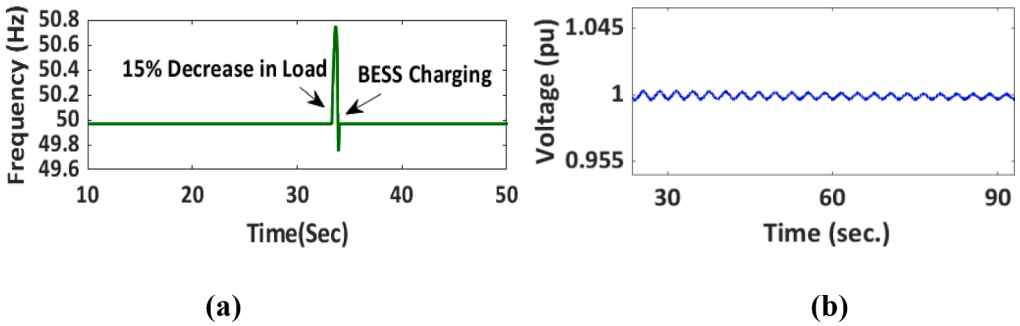


Fig. 12. During 15% decrease in load (a) Frequency (b) Voltage.

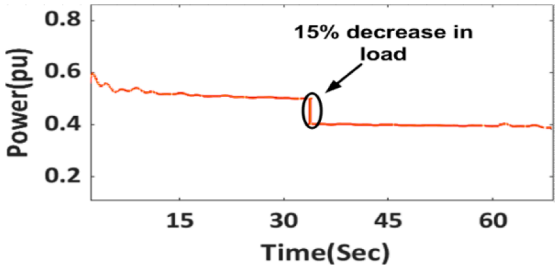


Fig. 13. Load power at bus 645.

resistance and q_{source} is the heat source. The sign will be negative for cooling loads, while for heating loads, it will be positive. The inside facing of the wall can be assumed having energy $q_x = \mp kA\frac{dT}{dx}$dT. The temperature difference for a small length dx. Then, the outside facing of the wall assumed to be having an energy of:

$$q_{x+dx} = \mp kA\left(\frac{dT}{dx} + \frac{d^2T}{dx^2}dx\right) \tag{15}$$

Taking $\Re_{wall}$ as a function of the perimeter P, differential length dx, and thermal capacity per unit area h, the energy lost due to convection is given as:

$$q_{flow} = \mp hPdx(T_{in} - T_{out}) \tag{16}$$

Similarly, for cooling loads, heat entered into the room is taken in terms of q_{flow}, the sign would be positive. If the room's heat storage, excluding any heat source, then from the dynamic modeling of TCL point of view.

$$q_{source} = hPdx(T_{out} - T_{in}) + C_{room}\frac{dT}{dt} \tag{17}$$

Where, C_{room} is outlined as the room's thermal capacitance illustrated in terms of kilowatt-hour/ °C. It is to be noted that the slope dT/dt may be negative or positive for cooling loads or heating loads, respectively, and the direction of heat-flow will be reversed as per the nature of the TCL. The geysers/air-conditioner has been denoted as a controlled current source. The resistance of the wall R_{wall}, in which heat is dissipated through conduction and convection (assuming the room has air ventilators), resulting in a temperature difference. $T_{in} - T_{out}$. A second-order equation can be derived by combining Eqs. (15) and (16) and writing the temperature difference as T which is given as:

$$q_{source} = kA\frac{d^2T}{dx^2}dx - hPTdx \tag{18}$$

TCLs, DZs, and battery storage devices are aligned to act in such a manner that their synchronization controls the frequency when it falls below the estimated value (50 Hz).

The percentage of TCLs to be essentially operated for frequency regulation depicted as:

$$\%TCL_{t,\ ON} = \%TCL_{t-1,\ ON} + M\Delta f \tag{19}$$

$$\%TCL_{t,\ OFF} = 100 - \%TCL_{t,\ ON} \tag{20}$$

Where, M depicts the aggressiveness of the TCL manipulation during frequency regulation. Additionally, an algorithmic model with a dead-

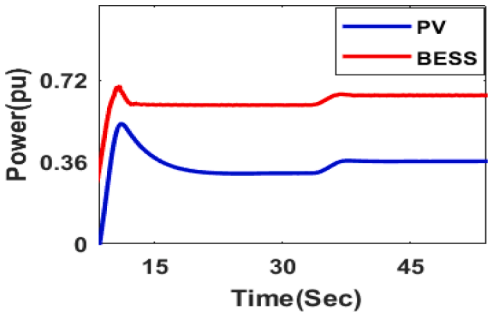


Fig. 15. DGs power during overcast conditions.

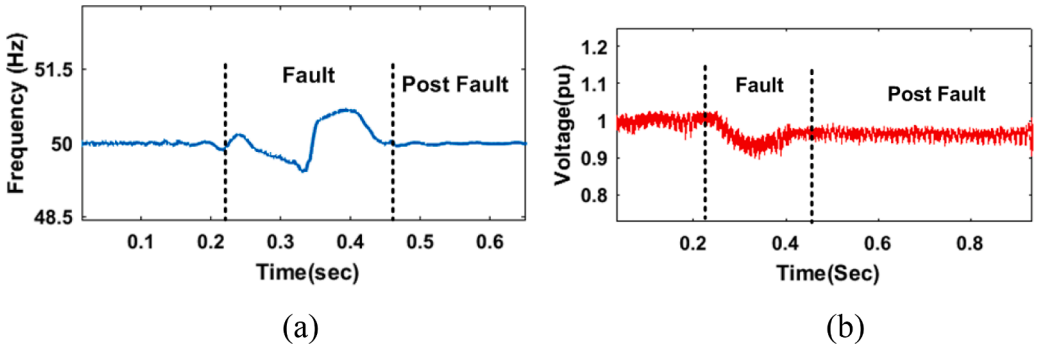


Fig. 14. During faulty condition (a) Frequency (b) Voltage.

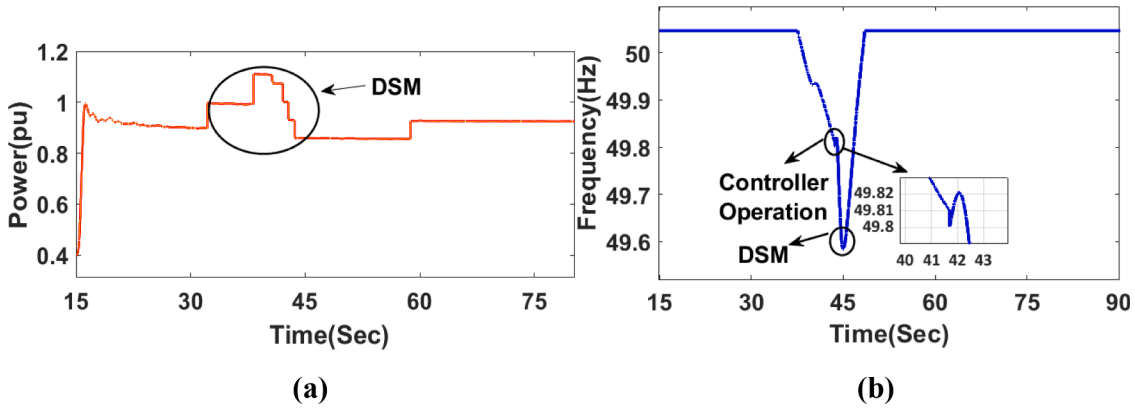


Fig. 16. (a) Different loading condition (b) Frequency response during peak hour.

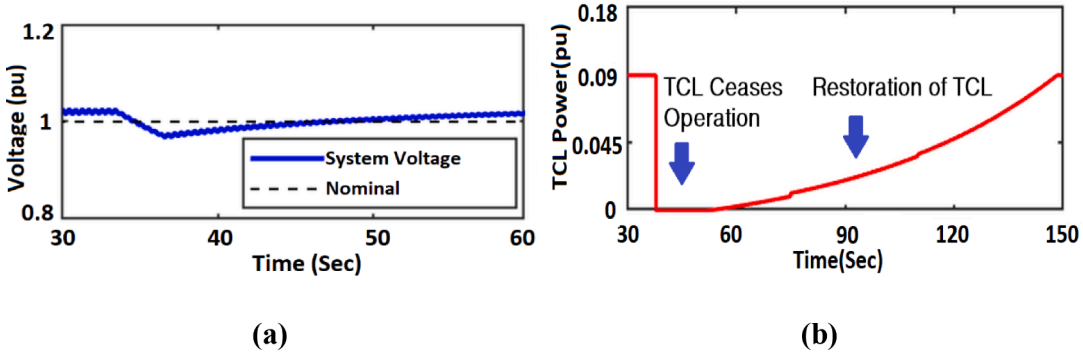


Fig. 17. (a)Voltage during peak hours at bus 645 (b) TCLs power participated for DSM.

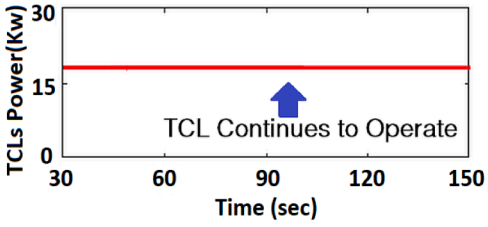


Fig. 18. TCLs power not participated in DSM.

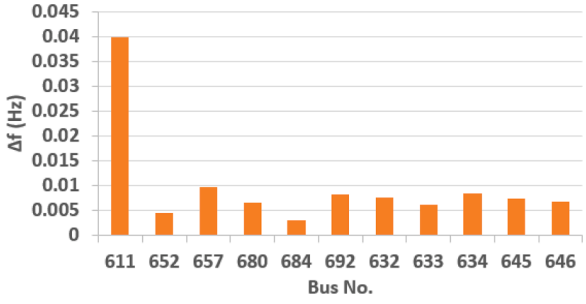


Fig. 19. Deviation in frequency at various buses after load management.

zone of 50 Hz ($f_{nominal}$) +0.6% to −0.6% has been implemented for droop characteristics. The droop characteristics of P-Δf can be expressed using the following equation:

$$P_{geyser} = P_{geyser,ref} + M_p \Delta f \quad (21)$$

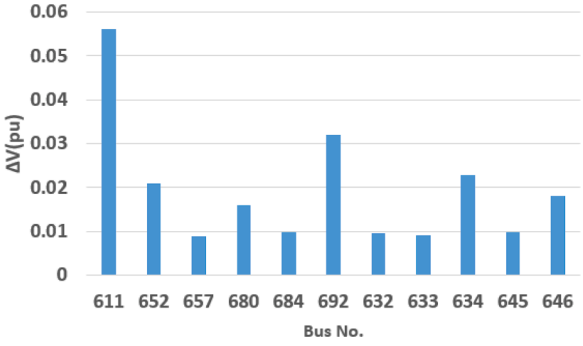


Fig. 20. Deviation in voltage at various buses after load management.

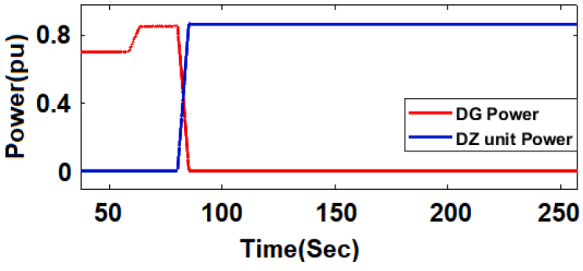


Fig. 21. DZ power during emergency condition.

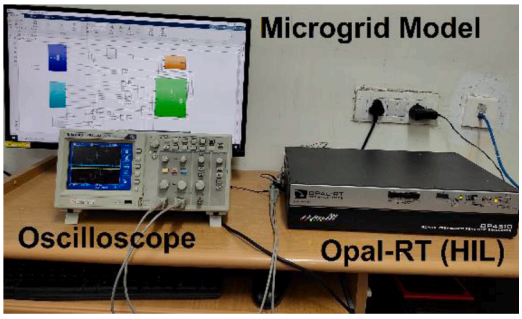


Fig. 22. HIL experimental setup.

$$P_{aircon} = P_{aircon,ref} + M_p \Delta f \tag{22}$$

P_{aircon}: Power consumption of the air conditioner in watts, $P_{aircon,ref}$: Air conditioner power consumption reference in watts, P_{geyser}: Power consumption of the geyser in watts, $P_{geyser,ref}$: Geyser power consumption reference in watts., Δf: Frequency deviation from its nominal value. M_p: frequency deviation associated with the power consumption coefficient. (W/Hz).

The TCLs (both heating and cooling loads) present in the studied system consumes 25% of total load power. The test case assumed to be in temperate zone. The deep learning- based technique is applied for TCLs aggregation, segregation based their type and temperature band, and followed by priority-based control. It starts with generation of data set at various operating condition, that includes operating temperature, frequency and state by monitoring customer's energy consumption pattern. The generated data (5000 cases) set with three variables are set against an output of either class:1(16–22 °C), or class:2(40–50 °C), or class:3(off state, on state but not in desired operating temperature band). 80% of the generated data set fed to a 5 layer-based CNN (Convolution Neural Network) to train the classification model. The rest 20%data used to test the performance accuracy of the trained model and it is found 99.06% accurate.

The complex CNN is done in hidden layer to determine the relation between input and output. The proposed CNN model is shown in Fig. 3. The temperature band of TCLs to be scheduled are considered as 16–22 °C for cooling (Air Conditioner) and 40–50 °C for heating (Geyser) loads.

The dataset required for training the CNN model given as

$$A = \left\{ (x_i,\ y_i) : x_i \in R^{+}_{1\ \times\ (1+2g)},\ y_i \in \ \left\{ [i]^{+}_{1\ \times\ 3} \right\},\ i = \{1, \ldots\ldots\ldots\ldots\ldots \mathrm{K}\} \right\} \tag{23}$$

Where K denotes the total number of examples during training of model.

$$\begin{aligned} O^{(1)}_{1xh} &= f([i]_{1\times(1+2g)}[W^{(1)}]^{t}_{h\times(1+2g)}) \\ &\vdots \\ \widehat{n_t} &= O^{(l)}_{1\times 3} = f([O^{(l-1)}]_{1\times p}[W^{(l)}])^{t}_{3\times p}) \end{aligned} \tag{24}$$

Where h = number of neurons present in 2nd layer, p = Neurons present in penultimate layer, l = Position of final layer, f = activation function (sigmoid), W = weight matrix, [i] = input matrix, n_t= operating state of 'g' TCLs in time 't'.

The objective function is formulated as follows,

$$\in_t = n_t - \widehat{n}_t \tag{25}$$

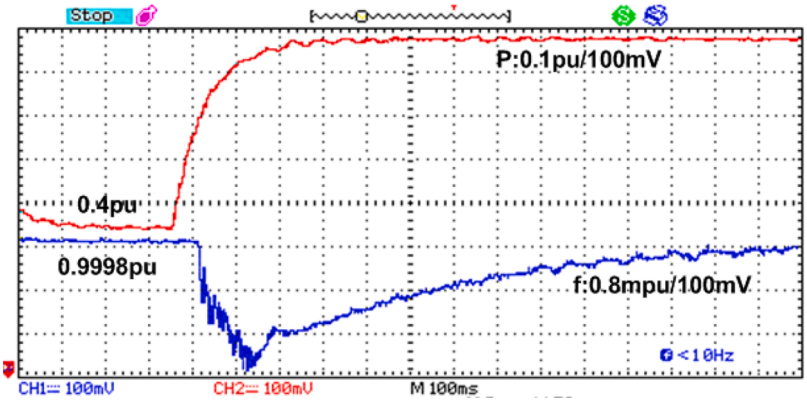


Fig. 23. Changing frequency with change in loading condition.

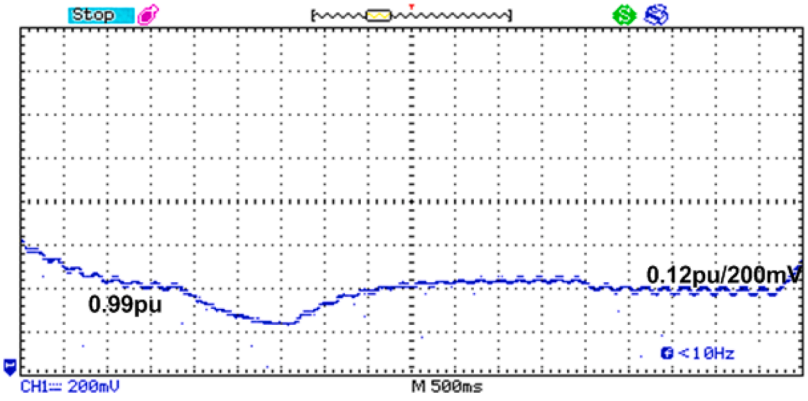


Fig. 24. RMS voltage waveform.

$$L(n_t, \hat{n}_t) = \frac{1}{N}\Sigma_{i=1}^{K}\epsilon_t^2 \tag{26}$$

Where $L(n_t, \hat{n}_t)$ is the loss function, to minimize L, calculating gradient

$$\frac{\partial L}{\partial W_{ij,iter}^{(z)}} = \frac{\partial L}{\partial O_{1\times 3}^{(l)}} \frac{\partial O_{1\times 3}^{(l)}}{\partial O_{p\times 1}^{(l-1)}} \ldots\ldots \frac{\partial O_{q\times 1}^{(z-1)}}{\partial W_{ij,iter}^{(z)}} \tag{27}$$

Where z is the position of layer where weight to be updated, q and p are the output matrix of hidden layers.

Updating the weight matrix

$$W_{ij,iter+1}^{(z)} = W_{ij,iter}^{(z)} - \alpha\, mom_{iter} \tag{28}$$

$$mom_{iter} = \beta\, mom_{iter-1} + (1-\beta)\frac{\partial L}{\partial W_{ij,iter}^{(z)}} \tag{29}$$

Where α and β are the learning rate ($\alpha = 3500$, $\beta = 0.89$) and moving average hyperparameter and mom is the momentum value momentum value for iteration 'iter' [28]. The Adam optimizer is used with a mean square error as a loss function. The classifier aggregation task is performed at the beginning to aggregate large population of TCLs followed by segregating as heating and cooling loads within the schedulable temperature band.

Finally, the control signal will send to the TCLs within schedulable temperature band (class 1 & 2) considering heat flow rate and priority based on weather condition (in winter ACs to be managed first then geyser as per requirement and vice versa during summer). Table 3 depicts the architecture specification of proposed model. The performance of the implemented model where the decrement in loss and increase of accuracy is shown in Fig. 4.

To evaluate the efficacy of the proposed model (CNN), the comparison is done with Support Vector Machine (SVM [29]) and k-Nearest Neighbours (KNN [30]). The classification performance of the proposed method (CNN) is found superior with 99.06% accuracy and is depicted in Table 4.

The flowchart depicting the primary and secondary controller operation is shown in Fig. 5, where the voltage and frequency is monitored and the appropriate action taken by the hierarchical controller both during peak and off-peak hours.

The coordination among the primary and secondary control strategy restrict the power ramp during load management. The % of TCLs take part in load management is decided by Eqs. (19) and (20). However, the after the load management the small deviation in voltage and frequency will be mitigated through primary control strategy.

4. Results and discussions

The hierarchical control strategy able to maintain voltage and frequency during peak and off-peak hours while considering several dynamics. The off-peak hours is simulated considering generation $\geq$ demand while during peak hours generation $<$ demand. The droop-based control strategy among PV and BESS responsible for power sharing. Fig. 6 shows the power sharing by the micro-sources (PV and BESS) during low irradiance with 1MW load. The performance of control strategy is validated considering system dynamics in following cases.

Case1: Sudden increase of 10% load at bus 645 (off-peak hours): This case is simulated considering the DGs are sufficient to deliver the load demand (generation $>$ demand). The off-peak hours abnormalities can be controlled through primary controller (Inverter Control). The load power variation with 10% increase in load is shown in Fig. 7.

The harmonic analysis is also performed and it is found that the total harmonic distortions (THDs) are within limit as per IEEE Standard 519–1992 and IEC Standard 61,000-3-6. Fig. 8 shows the voltage and current THD. The voltage THD is 1.73%, while the current THD is 2.72%. The voltage and current waveforms contain traces of higher-order harmonics.

The transient occurs due to sudden load switching is handled by the dual loop V-f control strategy and maintain the stability. A comparative frequency analysis is performed with droop-based controller given in Fig. 9 and proposed controller given in Fig. 10, where it is observed with droop-based control mechanism, the frequency reached below 48 Hz due to its slow response, while with inertia control strategy that RoCoF is less during transient.

Thus, the inclusion of virtual inertia concept with existing controller enhances robustness of the system. The corresponding RMS voltage is shown in Fig. 11, the voltage deviation is within limit. It can be observed that inverter controller is adequate in maintaining balance between generation and demand at load switching during off-peak hour (generation $\geq$ demand).

Case 2: Sudden removal of 15% load at bus 645 during sufficient irradiance (off-peak hours): The frequency and voltage response during high irradiance condition, while 15% load is suddenly removed from Bus 645 is shown in Fig. 12(a), (b). The load power at 645 is given in Fig. 13. However, this over frequency arises due to removal of load can be maintained through battery charging. Further, the primary controller is responsible to maintain the voltage and frequency. The inverter control operation is valid till generation and demand mismatch is small and the frequency is between 49.7 Hz and 50 Hz.

Case 3: Line to Line fault between 632 and 633 (off-peak hours): The performance of control strategy is also tested with LL fault. The case is simulated to validate the control strategy for enhancing microgrid resiliency. During off-peak loading (generation $\geq$ demand) of 800 kW a LL fault occurs between bus 632 and 633. The frequency and voltage response are shown in Fig. 14(a), (b). The fault is cleared around 0.35 s and the controller efficacy in restoring the nominal condition after fault is cleared is promising. During Peak hours or while demand is more (leading frequency beyond 49.7 Hz), the secondary control (load management) work with primary control to maintain stability.

Case 4: Sudden Integration of 20% load at bus 652 (peak hours): During peak hour or deficit generation or overcast condition or while the generation and demand mismatch leading frequency beyond 49.7 Hz, the DSM is implemented by manipulating TCLs through priority based deep learning-based technique. The DGs power (PV and BESS) is shown in Fig. 15. The load power profile during peak hour (generation and demand mismatch is high) is shown in Fig. 16(a) and (b) shows the frequency response. Further, the voltage response during peak loading condition is shown in and Fig. 17(a). It can be observed from Fig. 16(b), the controller fails to compensate the frequency variation and may makes the system unstable.

Thus, load (TCLs) management is carried out and the frequency reaches to nominal value. The deep learning based aggregated TCLs with low heat flow rate are considered for management. Thus, not to compromise the CTC. Power variation of the TCLs participating for frequency restoration is shown in Fig. 17(b). The TCLs within the temperature band and having low heat flow rate are cease to operate till frequency reached to the nominal value(50 Hz). The load management will be implemented as the frequency falls below 49.7 Hz.

The power of non-participated TCLs is shown in Fig. 18 (Due to high heat flow rate or outside the temperature band). The change in frequency (Δf) and voltage (ΔV) in all the busses after implementing DSM is shown in Figs. 19 and 20 respectively. It is observed that the voltage and frequency lie within the acceptable range adhere to IEEE 1547.

The PV, BESS, TCLs coordinate with each other to maintain the system power quality. However further increase in load degrade the system performance and required dynamic inertia (emergency condition) hence DZ will operate to maintain stability.

Case 5: During Emergency Condition: This case is simulated when PV generation is absent with low battery SoC. The DZ will deliver the load and avoid load shedding when the combined effect of inverter control action and load management fails to mitigate the generation and demand mismatch. Fig. 21 shows the DZ power during such critical

condition (emergency condition).

5. Real time validation results

The real time validation is done considering load dynamics to showcase the efficacy of the control strategy. The OPAAL-RT 4510 (HIL) is taken for real time validation with 5μsec sampling rate. The Simulink model is updated using RT LAB software, considering the power system and monitoring system. The technical specifical of the proposed model is given in Table 1. The twin channel Digital oscilloscope is used for signal extraction and monitoring purpose. Fig. 22 shows the picture of the HIL setup. The scenario is validated with increase in load at bus at bus 645. The load power vs frequency plot is shown in Fig. 23 considering generation > demand. It is observed that with sudden change in load the under frequency occur hence primary control strategy responsible to maintain the nominal condition adhere to IEC 62,109. Further the corresponding RMS voltage waveform is shown in Fig. 24.

6. Conclusion

The paper proposes a hierarchical voltage and frequency control considering both generation and end-user control. The primary control strategy of dual loop inverter fed droop-based voltage and frequency control with inertia support found to be adequate while frequency lies between 49.7 Hz and 50 Hz (off peak hours). Further, limit the overshot occurs during transient events and also maintain the rated voltage of 4.16 kV. However, while frequency falling below 49.7 Hz load management is adopted, comprise of 25% TCLs of total load to restore frequency adhering to IEC 62,109 standard. The coordination among inverter controller (primary control) and load management (secondary controller) manages the power quality during peak and off-peak hours. Further during peak hours Δf observed to be -0.6% and ΔV is 2% which validate the efficacy of the control strategy. Further, the priority based deep learning TCLs management technique, enable easy selection of TCLs to be controlled (considering weather condition and CTC) and provide auto-updating facility in regular interval which is found to be 99.06% accurate.

CRediT authorship contribution statement

Sourav Chakraborty: Writing – original draft, Investigation, Methodology, Software, Visualization. **Susmita Kar:** Conceptualization, Methodology, Writing – review & editing, Supervision. **Subhransu Ranjan Samantaray:** Writing – review & editing, Methodology, Visualization, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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